

GAVAGAI

Cross-situational grounding as balanced optimal transport

A referential-alignment objective for infant-scale vision-language models, targeting the BabyVLM Workshop @ NeurIPS 2026.

In one paragraph

A baby hears **“look — are you hungry? there’s the ball!”** while facing a room containing a ball, a hand, the floor and a table. Which word goes with which thing? Nobody says. Today’s baby vision-language models are trained to predict the next word given the picture, an objective with **no place in it for a word to mean a thing**. This work adds that missing piece: on every scene, words are assigned to image regions by solving a small transport problem with two rules — a word may refer to **nothing**, and no single region may absorb **everything**. The second rule is mutual exclusivity, a documented bias in real toddlers, and it turns out to be exactly a capacity constraint on an assignment matrix.

What this report covers

Section 1 states the problem. Section 2 summarises the papers the workshop asks entrants to read and what each leaves open. Section 3 identifies two gaps — one in how these models are **evaluated**, one in how they are **trained**. Section 4 explains the method without mathematics. Sections 5–6 give the novelty claim and the measured results. Sections 7–8 record what went wrong and what remains unproven.

1. The problem: which word means which thing?

The philosopher W.V.O. Quine imagined a linguist who hears a native speaker say **“gavagai”** as a rabbit runs past. Does it mean rabbit? white? running? dinner tonight? A single scene cannot tell you. An infant faces this problem continuously, and solves it.

Two properties of real infant head-camera footage make it considerably harder than the textbook version:

- **Most words refer to nothing on screen.** “hungry”, “there”, “we”, “go” point at nothing visible. In transcribed caregiver speech these are the majority of word tokens.
- **A few objects are in almost every frame and are almost never named.** Hands, floor, table, the caregiver’s torso. They are permanently available to be blamed for any word — the raw material of what we call a **hub**.

Developmental psychology names two mechanisms infants use. **Cross-situational learning**: no single scene resolves “ball”, but across hundreds of scenes the word co-occurs with balls and not with anything else. **Mutual exclusivity**: shown a familiar ball and a novel gadget and told “find the dax”, a toddler picks the gadget — one word, one thing. Both are the design targets of the method in Section 4.

2. What the prior work established, and what it left open

These are the papers the BabyVLM workshop lists as recommended reading, plus two that turned out to matter.

Paper	What it contributed	What it left open
SAYCam Sullivan et al., 2022	Head-camera video from three infants, 6–32 months, ~478 h. The reference corpus.	Gated behind Databrary approval, so results on it are not directly reproducible.
BabyView Long et al., 2024	A second, higher-resolution longitudinal infant corpus.	Also gated.
LLaVA Liu et al., 2023	The standard recipe: frozen vision encoder, connector, language model, trained to predict the caption.	Nothing in the objective ever asks which word corresponds to which part of the image.
CVCL Vong et al., Science 2024	Trained on one child’s footage; genuinely acquired word-referent mappings that generalise.	Matches whole utterance to whole frame. No latent word-to-region variable.
BabyVLM-V1 / V2 Wang et al., ICCV’25 / CVPR’26	A 1.1B model trained from scratch on SAYCam, plus DevCV Toolbox : ten tasks adapted from the NIH Baby Toolbox®.	Every training stage is next-token prediction. No contrastive term anywhere.
Looking to Learn Ganesu et al., 2025	A learned gate deciding when to use vision versus language. Found it favours vision for content words, language for function words — with no supervision.	Left as an emergent side effect rather than a structural commitment.
3rd BabyLM Challenge Charpentier et al., 2025	Ran the multimodal track with GIT and Flamingo baselines.	No submission beat the baselines, two years running. Notes that new objectives tend to produce the best approaches.

3. Two gaps

3.1 The benchmark may not measure what it appears to

BabyVLM-V2’s published per-task scores look like a model with excellent perception and no vocabulary: **96.4** on Left/Right (humans score 94.5 — it beats people) but **32.4** on Picture Vocabulary, where pure guessing scores 25.0 and humans score 91.8.

But Left/Right asks: “here is a cat — which of these three is exactly this cat?” That is a spot-the-matching-photograph puzzle. It may not require understanding the question at all. So we tested it with a deliberately weak baseline: shrink every image to 32×32, take its brightness pattern and colour histogram, and return whichever option is most similar to the query. **No training, no neural network, and no access to the prompt.**

Task	Text-blind matcher	Baby model	Human	Chance
Left/Right	1.00 (24/24, CI 0.86–1.00)	96.4	94.5	33.3
NIH Spatial	1.00 (10/10, CI 0.72–1.00)	92.8	100	33.3
spatialdetails (SAYCam)	0.33 (CI 0.10–0.70)	—	—	33.3
Picture Vocabulary	n/a — nothing to match against	32.4	91.8	25.0

Finding 1

Left/Right — where the published model scores **above the human ceiling** — is solved outright by a matcher that never reads the question. Picture Vocabulary is **not**, which is the control showing the baseline is not simply winning everywhere. Part of the dramatic “perception yes, words no” split is therefore a property of the **tasks**, not of the model. **Caveat:** n is 6–24 per task on the public samples, hence the Wilson intervals; these must be regenerated on the full public Ego4D release before being quoted.

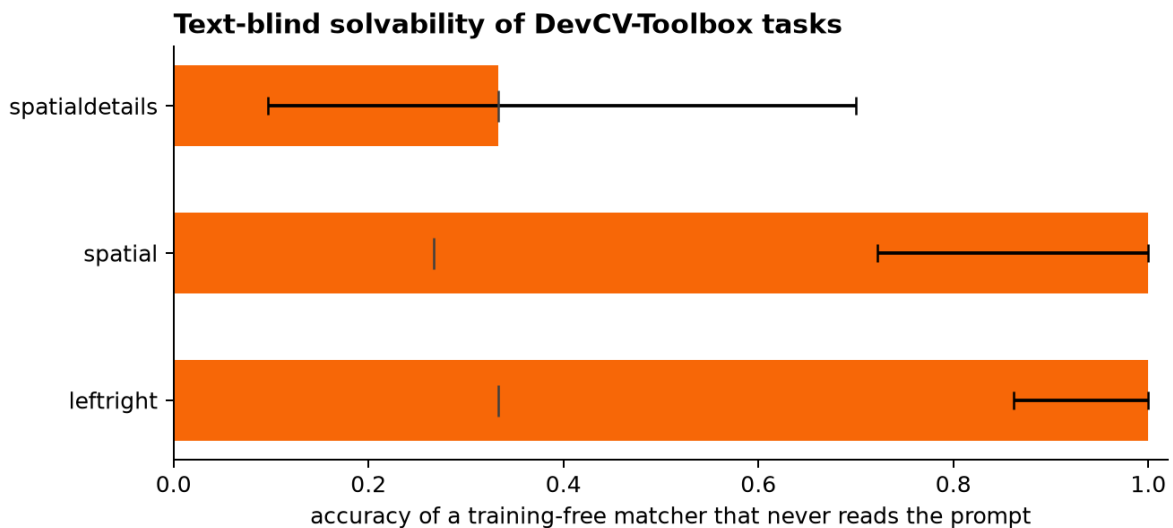


Figure 1. Accuracy of a training-free matcher that never reads the prompt. Vertical ticks mark the chance floor for each task.

3.2 The training objective has nowhere to put a word’s meaning

This is the deeper gap. In LLaVA-style training the model’s only job is: given the picture, predict the next word. That can be done well from language habits alone — “are you...” is usually followed by “hungry” — plus a rough sense of the scene. There is no variable anywhere in the mathematics that says **“this word refers to that region”**, so there is no pressure ever to work it out.

A note on precision: it is tempting to blame contrastive learning’s frequency bias here. That would be wrong, and we do not claim it — with negatives drawn from the marginal, the contrastive optimum is already an unbiased association score. The defect is specifically in the **within-scene** assignment step, whose competitors are the other regions of the same image.

4. The method, as a seating chart

Treat every scene as a seating problem. The rows are the content words just heard. The columns are regions of the image — plus one extra column labelled **NOTHING**. Every word must be assigned somewhere.

	hand	floor	ball	table	NOTHING
ball			■		
hungry					■

Figure 2. One scene as an assignment matrix. “ball” claims a region; “hungry” claims the NOTHING column.

Rule 1 — a word may mean nothing

That is what the NOTHING column is for. This sounds trivial; it is the single most important part. Without it, “hungry” is forcibly glued to whatever happens to be visible — usually a hand or the floor, because those are always there. Repeat a few thousand times and every word in the vocabulary ends up pointing at a hand.

Rule 2 — no region may absorb everything

Each column has limited capacity. If “hand” has already been claimed, the next word must look elsewhere. This **is** mutual exclusivity, expressed as a constraint rather than a heuristic: the degenerate solution where every word points at one salient region becomes **infeasible**, not merely disfavoured.

Solving it

“Assign rows to columns under capacity limits” is a classical, solved problem — **optimal transport** — and the standard algorithm is simply: repeatedly divide each row by its total, then each column by its total. About eight matrix multiplications. Next to running the vision encoder it is effectively free.

Remembering across scenes

One scene is hopeless; a thousand are not. A running tally records how often each word was assigned to each kind of object, and that tally is fed back into the next scene’s costs. Crucially the tally is scored by **how surprising** a pairing is, not how frequent — raw counts would just conclude that everything goes with hands, since hands are always present. Correcting for baseline frequency is what lets “ball” stand out. This is the cross-situational learning the project is named for.

5. What is actually new

Ingredient	Why it matters
One dial from the standard method	A single knob p controls exclusivity. At $p = 0$ the mathematics provably collapses into exactly the region-word contrastive learning everyone already uses. Every ablation is therefore the same code path with one number changed — no chance of accidentally handicapping the baseline through a re-implementation.
Words may refer to nothing	Standard objectives force every word onto some region. Given that most caregiver speech is non-referential, this is a genuine defect; no prior work makes “nothing” a first-class option.
Memory across scenes	The closest prior art (OTTER, 2021) uses related transport machinery, but only within a single batch and at whole-caption granularity. Here the coupling is word-to-region and the lexicon persists across the entire run.
Exclusivity as a constraint	A bias psychologists have studied for four decades turns out to be precisely a capacity constraint on an assignment matrix — and enforcing it costs almost nothing.

6. Measured results

All figures below are generated directly from the run outputs; no number is transcribed by hand. Every condition is three random seeds with dispersion reported.

6.1 The headline: forcing every word to mean something is catastrophic

Picture-vocabulary accuracy on **held-out exemplars** of each category — so this measures generalisation to unseen instances, not recall of a memorised vector. Chance is 0.025.

Condition	Clean	Moderate	Realistic
Captioning only (the standard recipe)	0.392	0.242	0.058
+ alignment, forced (no NOTHING column)	0.650	0.358	0.000
+ NOTHING column only	0.658	0.358	0.017
+ NOTHING + exclusivity (ours)	0.700	0.400	0.075
+ NOTHING + full balancing	0.683	0.400	0.083

Finding 2

Under realistic ambiguity — 80% of speech non-referential, half of content words uttered while their referent is off-screen — the **naïve** form of the alignment scores **0.000, with zero variance across all three seeds**. That is total hub collapse, and it is **worse than adding nothing at all**. The escape hatch and the capacity limit are what turn that catastrophe into the best result in the table.

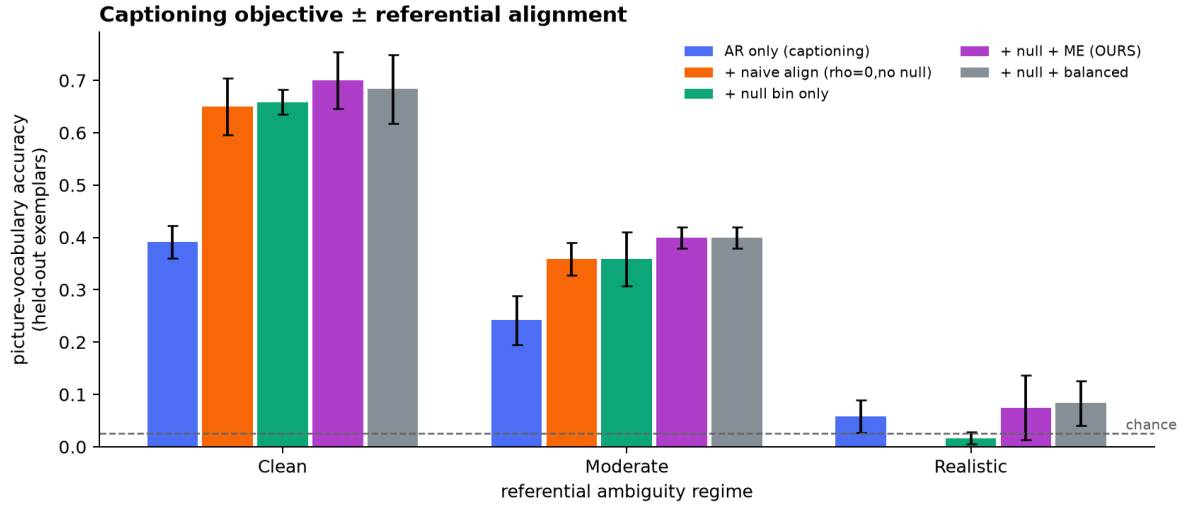


Figure 3. The captioning objective is weakest in every regime. The missing orange bar at right is not a rendering error: naive alignment scores exactly zero.

6.2 The exclusivity knob has a sweet spot

Sweeping ρ continuously from 0 (which is the standard method) to full balancing:

ρ	0	0.02	0.05	0.1	0.3	1	3	∞
accuracy	0.275	0.333	0.375	0.383	0.358	0.333	0.358	0.350
hub rate	0.050	0.033	0.050	0.033	0.042	0.025	0.025	0.025

The optimum is **interior**, not at maximum exclusivity. That is what one wants from a genuine cognitive constraint: real scenes do contain several things a word could plausibly mean, so forcing a strict one-to-one matching over-constrains. A parameter that only ever helps would look more like an artefact than a mechanism.

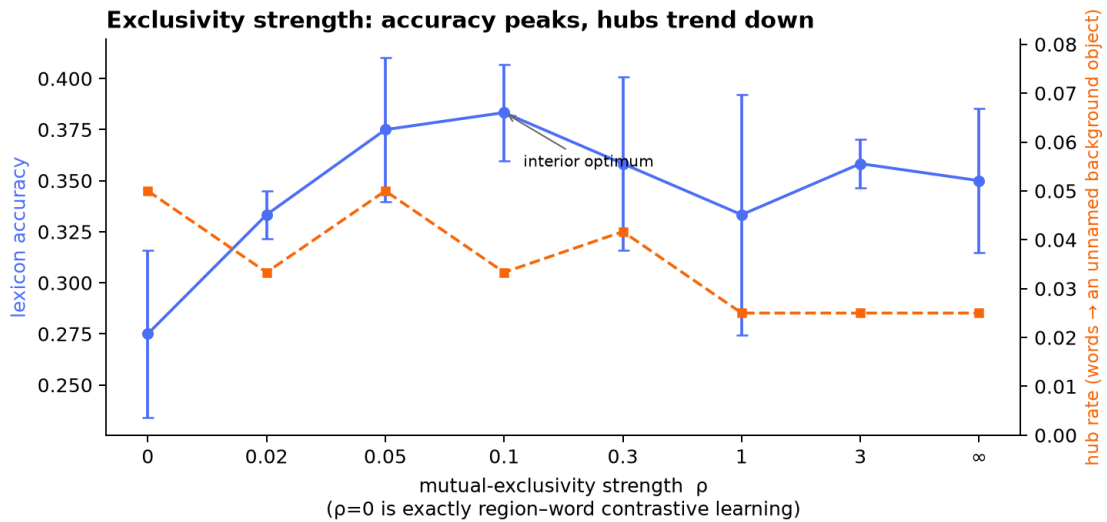


Figure 4. Accuracy peaks at an interior optimum while the hub rate trends down — the effect and the mechanism it is attributed to move together.

6.3 Consistent across corpus sizes

Re-measured sample efficiency; column A is the standard method ($\rho = 0$, no NOTHING column):

Corpus	A — contrastive	B — + NOTHING	C — + exclusivity	D — balanced
250	0.342	0.333	0.400	0.383
500	0.358	0.417	0.458	0.475
1000	0.392	0.367	0.475	0.467
2500	0.450	0.350	0.542	0.492

Condition C beats the baseline at all four sizes (+0.058, +0.100, +0.083, +0.092), with hub rate roughly a third to a quarter of the baseline’s throughout. **The accuracy gain and the mechanism it is attributed to move together** — which is stronger evidence than a bare accuracy delta.

6.4 It reproduces human data

Yu & Smith (2007) had adults learn 18 word-referent pairs from 27 deliberately ambiguous trials. An unconstrained ideal observer solves that design perfectly, so the informative question is what capacity limit reproduces the human pattern. Forgetting does not — accuracy stays at ceiling for every decay rate we tried. **Limited encoding** does: assume a learner registers only two word-object pairs per trial, and one shared free parameter reproduces the whole ordering.

Objects encoded per trial	2×2	3×3	4×4	RMSE vs human
unlimited	0.983	1.000	1.000	0.292
1	0.456	0.336	0.208	0.410
2	0.983	0.725	0.517	0.066
3	0.983	1.000	0.875	0.231
human	0.889	0.778	0.556	—

7. What went wrong along the way

Recorded here rather than quietly patched, because each one would have been a reviewer’s finding.

What was wrong	How it was caught and fixed
The original theory was wrong. The first framing blamed contrastive learning’s frequency bias.	It is not frequency-biased — with marginal negatives its optimum is already an unbiased score. Reframed around the within-scene assignment step, which genuinely lacks any correction.
A tempting lemma was false. “Balancing induces the same ranking as the surprise score.”	Tested on 2,000 random matrices: the two disagree on more than half. There is now a test that documents the falsity so nobody re-derives it.
A negative result was retracted. An earlier draft reported the exclusivity constraint gave no benefit.	It had been measured in a setup where every condition scored a perfect 1.000, making the comparison vacuous. Re-measured properly, the effect is present at every corpus size.
Part of the codebase was silently untracked.	A .gitignore pattern matched at any depth and excluded a package directory. A fresh clone would not have imported. Caught by actually cloning it.
A performance defect that only bites on GPU.	The transport solver synchronised with the host every iteration — harmless on CPU, crippling on a T4. Removed before any GPU time was spent.
Tests depended on execution order.	Random seeding sat at module level, so adding a new test file silently changed the data another file’s tests ran on. Seeding is now per-test.
A retracted claim was baked into the generator.	The summary document is auto-generated; the withdrawn wording lived in the generating script and would have reappeared on every regeneration.

8. What remains unproven

- **Everything measured so far is simulation.** Whether the gains transfer to real video is untested, and is the genuine risk in the project.
- **The benchmark audit rests on small samples** (6–24 items per task). Regenerating it on the full public Ego4D release is the single highest-value next step — minutes of compute, and unlike the training run it cannot come back negative.
- **The hardest ambiguity regime is floor-limited.** Tripling the corpus and the training steps moves nothing, so no claim is made there beyond “does not collapse”.
- **Regions are pooled grid cells, not discovered objects.** No object discovery is claimed.
- **The public data is adult egocentric video** (Ego4D), used as a reproducible stand-in for gated infant corpora. Claims about developmental plausibility are hedged accordingly.

How the work is structured against risk

This is deliberately a **two-tier bet**. The benchmark audit in Section 3.1 is a measurement that holds up regardless of whether the method works — it needs no training and cannot fail. The method sits on top of it with supporting evidence and its limits stated. If the real-data run disappoints, the contribution narrows rather than disappears.

Reproducing everything

```
git clone https://github.com/Kylian07/Baby-VLM
pip install -r requirements.txt && python -m pytest tests/ -q
python scripts/text_blind_audit.py --roots data/Ego4D
python scripts/run_simulation.py ar
```

31 tests cover the objective, the transport solver, the lexicon and the benchmark loader — including one test whose purpose is to document a claim verified to be **false**. A Kaggle notebook runs the whole pipeline end to end on a T4.

The single highest-value next hour

Run the benchmark audit against the **full public Ego4D release** of DevCV Toolbox — one command in the Kaggle notebook, minutes of compute. It turns the Left/Right result from **24/24 with a wide confidence interval** into a number with enough samples to headline a paper. Unlike the training run, it cannot come back negative: whatever it returns is a fact about the benchmark that the community needs.

Repository: github.com/Kylian07/Baby-VLM · Method and proofs: [METHOD.md](#) · Measured numbers: [RESULTS.md](#) · Reading notes: [docs/related_work.md](#) · Paper plan: [docs/paper_outline.md](#)